

Adelowo Adebangi Oluwatimileyin

Software Engineer · Python Backend & ML Systems

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PROFILE

Software engineer with 3+ years of industry experience spanning ML applications, data-processing workflows, and operational analytics. Builds Python REST APIs and Dockerised dashboards that connect models and telemetry to internal users. Brings additional experience in computational geometry, numerical algorithms, and reproducible scientific software.

PROFESSIONAL EXPERIENCE

Data Scientist | *Intuos Srl*, L'Aquila, Italy

Dec 2024 – Present

- Build a Dockerised FastAPI / IBM DB2 backend and React/Vite dashboard for internal flight-performance analysis.
- Expose ML predictions and engineering metrics through REST APIs; integrate deterministic safety-alarm logic.
- Develop processing and analytics workflows over recorded, externally ingested aircraft telemetry.
- Implement FFT-based audio processing for engine RPM estimation and analyse engine-order harmonics on aircraft recordings.

Machine Learning Engineer | *N-and Italia Srl*, Italy

Apr 2023 – Nov 2024

- Developed predictive models for vending-machine delivery-failure forecasting using large-scale transaction and telemetry data.
- Performed exploratory and statistical analysis to identify patterns affecting performance and operational reliability.
- Built data-processing and KPI-monitoring workflows to support operational decision-making.

Operational Data Analyst | *eResult Srl*, Cesena, Italy

Jan 2023 – Mar 2023

- Optimised SQL queries and automated recurring business-intelligence reporting workflows.
- Transformed operational data into reproducible performance metrics and actionable reporting outputs.

Digital Engineering Intern | *Bridgestone EMIA*, Rome, Italy

Apr 2022 – Oct 2022

- Developed computational-geometry algorithms and improved data-structure operations for tire cross-section simulation and finite-element meshing workflows.

Machine Learning Intern | *Fellowship.AI*, Remote

May 2022 – Aug 2022

- Implemented an InsetGAN architecture for whole-body image generation and deployed model inference through FastAPI for integration with an iOS application.

TECHNICAL SKILLS

Backend & data: Python, FastAPI, REST APIs, SQL, PostgreSQL, IBM DB2

Applications & tooling: React/Vite, Docker, Git, MLflow; telemetry and model-serving workflows

Algorithms & computing: C++, MATLAB, NumPy, SciPy, Pandas; computational geometry, FFTs, numerical methods

ML libraries: PyTorch, scikit-learn, TensorFlow, LightGBM

SELECTED PROJECTS

Flight Data Monitoring Dashboard

- Built a FastAPI / IBM DB2 backend and React/Vite frontend, packaged with Docker, for internal analysis of recorded aircraft telemetry.
- Integrated flight-phase classifiers, engineering metrics, and deterministic safety alarms into a query-on-demand dashboard over recorded telemetry.

Aircraft Engine Monitoring via Audio DSP

- Implemented microphone-based FFT processing on Raspberry Pi, transmitting RPM and status over UART to an ESP32 display controller.
- Evaluated the signal-processing pipeline on Tecnam P92 / Rotax 912 audio recordings and characterised engine-order harmonics.

LLM Applications and Model Compression

- Implemented medical retrieval-augmented generation with LangChain/ChromaDB and a two-stage enterprise natural-language-to-SQL pipeline.
- Implemented FinBERT layer removal and knowledge distillation, reducing the model from 110M to 88M parameters.

Verified Numerical Simulation Software

- Built a Fourier pseudo-spectral Navier–Stokes solver and tensor reduced-order model; verified temporal and spatial convergence against analytical and manufactured solutions.
- At rank 16, achieved $38.5\times$ online speed-up with $< 0.1\%$ mean state error on the training trajectory.

INDEPENDENT RESEARCH PROJECTS

Applied Mathematics & Scientific Machine Learning

Jan 2026 – Present

Develop and verify numerical solvers and ML surrogates using convergence studies, analytical solutions, gradient checks, and held-out experiments. Study model accuracy, runtime, and generalisation in PDE simulation and computational imaging.

EDUCATION

M.Sc. Mathematical Engineering | University of L'Aquila, Italy

2018 – 2021

Thesis: *Bounds on Mixing of Passive Scalars Advected by Incompressible Energy- and Enstrophy-Constrained Flows and Anomalous Dissipation.*

Relevant study: machine learning, scientific computing, optimisation, numerical analysis, finite-element methods, and PDEs.

B.Sc. Mathematics | Obafemi Awolowo University, Nigeria

2013 – 2018

Thesis: *Application of Galerkin Finite Element Method to the One-Dimensional Stefan Problem.*

CERTIFICATIONS

IBM Data Science Professional Certificate (IBM / Coursera, 2020); Machine Learning (Stanford University / Coursera, 2020).

SELECTED AWARDS

Professor S.S. Okoya Prize for Excellence in Applied Mathematics (2018); University of L'Aquila International Scholarship (2018–2021).